GitHub PR Plan — MINERVA QCELL/RFCELL Colour-Line Model

Branch & PR strategy. **Consolidated** — we do NOT open one PR per wave. Completed waves are bundled into a single reviewable PR so the reviewer sees a coherent capability set, not a trickle of partial commits. **PRs are never auto-merged.**

PR1 — W001 + W002 + W003 + W004: Colour-line model, layer hierarchy & geometric tracing (THIS PR — ready for review)

Branch: `wave/w001-w004-colour-line-layer-geometry`
Base: `main`

This single PR consolidates the entire foundation: source ingestion + style extraction (W001), colour-line decomposition (W002), the 13-top-level / 21-named-sub-layer hierarchy (W003), and geometric arrow/flow tracing (W004).

Reproducibility (read first)

Derived outputs (`data/model/` , `data/pemo/` , `output_v6/` , `publish/` , `reports/*.xlsx`) are **git-ignored** — they are regenerable, not source. To reproduce every number quoted below from a fresh clone:

```
./make.sh                                         # regenerate all derived outputs
PYTHONPATH=src python3 tests/test_integration_pipeline.py   # source-only smoke test
PYTHONPATH=src python3 tests/test_colour_model.py           # W002 assertions (after make.sh)
PYTHONPATH=src python3 tests/test_w003_w004.py              # W003/W004 assertions (after make.sh)
```

Tracked source of record: `src/` , `segmentation/data/*.json` , `configs/` , `data/svg/` , `tests/` , `reports/*.md` , `docs/` .

PR description — Capability Matrix

Honest "Claim ≠ Complete" accounting. Every row is **CAN** (implemented & verified by runtime counts), **CANNOT** (not possible with current source / toolchain), or **DID NOT / DEFERRED** (possible but intentionally not done).

CAN — implemented and verified

Capability	Evidence
Ingest both real source SVGs with inline-style colour precedence	QCELL + RFCELL, 1888 drawable elements
Reduce unmapped elements via legend re-classification	982 → 112 (88.6 %)
CTM-resolved geometry extraction (bbox/ centroid/shape)	`geometry.py`; 7 shape classes
Pair text → components	315 pairs
Colour-classify text nodes	533 nodes
Pair dots / heat-load triangles / arrows → lines	205 / 100 / 132
Pairing-distance quality recorded	median 25.35 / p90 355.4 / max 1040.2 px (>p90 flagged low-confidence)
Standardize text to Consolas + 4 mm tiers	533 nodes tiered
Assign 13 top-level layers (21 named sub-layers) per element	`layer_assignment.json`
Layer sum-check balances exactly	1888 drawable + 533 text = 2421, no drops
Render layered SVG + PDF	QCELL 1834/14, RFCELL 591/12
Build interactive layer-toggle atlas	`layered_atlas_v6.html`
Trace flow arrows + junctions	132 arrows, 36 junctions
Parse vertical-letter nomenclature into tree	33 segments, 8 parents
Catalog components to Excel + HTML	297 components
Catalog spec-change dots by line	S 40, V 10, B 2, uncol. 153
Detect scope boundaries	5/5 (QM/QVB/vac/Jumper/QINFRA)
Detect handover diamonds (TP#NNN)	22
Emit PEMO YAML 1.2 SSOT	122 loops, 60 heat loads

CANNOT — blocked by source data / toolchain

Capability	Reason
Map the 112 residual magenta elements to a line	No legend swatch for that colour family in the source SVG
Confirm true 3-D pipe routing / elevations	Source is 2-D schematic; no z-data
Resolve the degenerate-transform coordinate outlier to a real position	Source transform is mathematically degenerate; mitigated via viewBox bounds

DID NOT / DEFERRED — possible, deferred to a later wave

Capability	Status / target wave
XLSX tag/instrument register reconciliation + coverage delta	**W005 (next, PR2)**
Recover the U line (top-left, currently black)	DEFERRED — colour re-attribution pass
Bind the 77 floating arrows to source lines	DEFERRED — nearest-line heuristic
Populate 04G_E_RED / manifold header layers	DEFERRED — reserved placeholders
Interactive layered viewer (beyond static atlas)	DEFERRED — W006, PR3
Cross-drawing reconciliation, temperature/ pressure annotation	DEFERRED — later waves
CI / GitHub Actions workflow	DEFERRED — W007, PR3

Review checklist

- [ ] `./make.sh` reproduces all derived outputs from a clean clone.
- [ ] Integration smoke test passes (`tests/test_integration_pipeline.py`).
- [ ] Confirm 13-top-level / 21-named-sub-layer assignment matches engineering intent.
- [ ] Sanity-check the 297-component catalog against the drawing.
- [ ] Confirm handover-diamond list (22) and 5 scope boundaries.
- [ ] Accept the documented deferrals (U line, 112 magenta, 77 floating arrows).

Not merged automatically

Opened for review only. Merge after sign-off.

PR2 — W005: Tag & Instrument Register Reconciliation (XLSX coverage delta) (planned — next)

This is the #1 next priority — it is the question an engineering reviewer asks first. Reconcile the 297 auto-catalogued components against the official tag/instrument register (XLSX) and report the **coverage delta**:
- which catalogued components appear in the official register (matched),
- which are **false positives** (catalogued but not in register),
- which register tags are **missing** from our catalog,
- nomenclature reconciliation (our segment labels vs official tag naming).

Cross-validates colour-derived line identity against ISA tag class (`configs/isa_classes.json`) per the governance principle in `AGENTS.md`.

PR3 — W006 + W007: Interactive layered viewer + CI (planned)

- **W006** — interactive layered viewer (beyond the static atlas HTML): toggle layers, isolate colour lines, click-through component inspection.
- **W007** — CI / GitHub Actions workflow running `./make.sh` + the test suite on every push.

PR4 — W008 + W009: Round-trip reassembly + publication & sign-off (planned)

- **W008** — recompose isolated colour lines back into the full P&ID without loss of meaning (the round-trip success metric).
- **W009** — engineering-reviewed colour-line atlas + sign-off record.

Conventions

- One consolidated feature branch per PR bundle; never commit directly to `main`.
- Commit identity: `Abacus Agent <agent@abacus.ai>`.
- Squash-merge with the wave ids in the title.
- Derived outputs are git-ignored and regenerated by `./make.sh` (see `.gitignore`).